

Mohi Arya

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology - Computer Science and Engineering

Vellore, Tamil Nadu, India

Aug 2023 - Expected 2027

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript

Web/Backend: React, Node.js, Express, FastAPI, HTML, CSS, Tailwind CSS

AI/ML: TensorFlow/Keras, OpenCV, Gemini API

Databases: PostgreSQL, Prisma, Turso

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Design & Analysis of Algorithms

Tools: Git/GitHub, Vite, Vitest, pytest, Vercel, Render

PROJECTS

Meal Finder & Planner — React, Vite, Node.js, Express, PostgreSQL, Prisma, JWT

[Live Demo](#) | [GitHub](#)

- Built a full-stack nutrition and meal-planning platform (React 19/Vite, Node.js/Express, PostgreSQL via Prisma) with JWT authentication, bcrypt password hashing, and per-user ownership checks enforced on every data-access endpoint.
- Designed a deterministic recommendation engine that filters a 110-recipe database against hard calorie, macro, dietary, and allergen constraints rather than relying on an LLM to get them right, with a local-first hybrid fallback to the Spoonacular API for recipes outside the local catalog.
- Covered the recommendation engine and cross-user data isolation with 99 backend and 3 frontend automated tests; deployed on Vercel, Render, and Neon (serverless PostgreSQL).

DSA Practice Coach — React, Node.js, Express, Prisma, Turso, Gemini API

[Live Demo](#) | [GitHub](#)

- Built a full-stack DSA practice tracker that schedules problem reviews using the SM-2 spaced-repetition algorithm (the same algorithm used by Anki), implemented as an independently unit-tested pure function.
- Integrated Google's Gemini API to generate Socratic hints that guide toward a solution without revealing code; enforced this with a post-hoc check on the model's actual output and an automatic retry, not just prompt instructions alone.

Deepfake Forensics Lab — Python, FastAPI, OpenCV, TensorFlow/Keras, React

[Live Demo](#) | [GitHub](#)

- Built an image-forensics tool that detects faces (OpenCV Haar Cascade) and classifies them as real or manipulated using MesoNet, a CNN trained on FaceForensics++, served via a FastAPI backend and React frontend.
- Ran an independent evaluation against a labeled FaceForensics++ dataset (separate from the model's own reference samples), computing real accuracy/precision/recall/F1 via scikit-learn (62.4% accuracy) rather than assuming performance, with a documented methodology covering dataset provenance and sampling rigor.

Social Media Content Analyzer — React, JavaScript, PDF.js, Tesseract.js, Vitest

[Live Demo](#) | [GitHub](#)

- Built a client-side React app that extracts text from an uploaded PDF or image and scores it for social-media engagement using a rule-based analysis engine, with no backend and no uploaded data leaving the browser.
- Implemented a dual-path text extraction pipeline: PDF.js reads the embedded text layer where available, with automatic per-page OCR fallback (Tesseract.js) for scanned or image-only pages.

CERTIFICATIONS

- Java Foundations Associate** — Oracle

ACTIVITIES

- QuantAlphaX - Quant Trading Workshop** — Bulls and Bears Club x WorldQuant Research (India), VIT (Apr 2024)
- Yantra'25 - Binary Battles 2.0** — Competitive coding event, International Society of Automation, VIT (Feb 2025), advanced to Round 2

LANGUAGES

Hindi: C2 English: C1